8. Generative Adversarial Network

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FS26 MTH345: Deep Generative Models: A Statistical Perspective





Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Likelihood Maximization Principle
4. Basics of Probabilistic Modelling

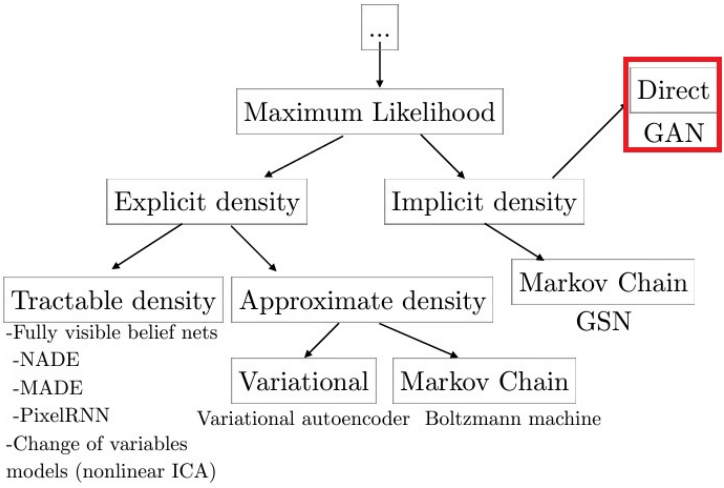
M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Lecture Topics



The figure is edited from Goodfellow (2016).

Lecture Topics

- Recommended reference materials include:
  - Goodfellow et al. (2014), "Generative adversarial nets," *Advances in neural information processing systems*.
  - Goodfellow (2016), "Nips 2016 tutorial: Generative adversarial networks," *arXiv*.

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Motivation

- Most previously reviewed methods, such as ICA, auto-regressive models, RBMs, and VAEs, explicitly formulate density models.
- However, due to non-linear generators, the likelihoods are often less tractable, which necessities the use of surrogate density forms.
- RBMs utilize the contrastive divergence algorithm:

$$\frac{\partial \log p_\theta(\boldsymbol{x})}{\partial \theta} = -\sum_{\boldsymbol{z}} \left(\frac{\partial E_\theta(\boldsymbol{z}, \boldsymbol{x})}{\partial \theta} \right) p_\theta(\boldsymbol{z}|\boldsymbol{x}) + \sum_{\boldsymbol{z}, \boldsymbol{x}} \left(\frac{\partial E_\theta(\boldsymbol{z}, \boldsymbol{x})}{\partial \theta} \right) p_\theta(\boldsymbol{z}, \boldsymbol{x}) \quad (1)$$

  where each term is approximated using a Markov chain.

- VAEs utilizes the variational inference:

$$\log p_\theta(\boldsymbol{x}) - \mathcal{D}_{\mathrm{KL}}(q_\phi(\boldsymbol{z}|\boldsymbol{x})||p_\theta(\boldsymbol{z}|\boldsymbol{x})). \quad (2)$$

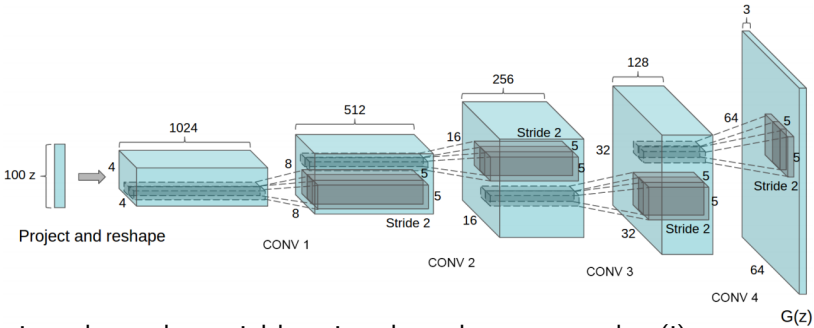
Motivation



- The approximations of (gradients of) log-likelihoods often fail, resulting in blurry generation results.
- Generative Adversarial Networks (GANs; Goodfellow et al., 2014) propose an alternative approach to approximating a likelihood ratio-based objective.

Images are from Tolstikhin et al. (2018).

Generative Adversarial Network



- GANs introduce adversarial learning through two networks: (i) a generator and (ii) a discriminator.
- The generator specifies the same graphical model used in VAEs, but $\boldsymbol{x} = G_\theta(\boldsymbol{z})$ where G is a neural network, i.e., $p_\theta(\boldsymbol{x}|\boldsymbol{z})$ degenerates to a single point.

Example of the generator architecture of deep convolutional GANs (Radford et al., 2016).

Adversarial Learning

- In adversarial learning, the key idea is to train a binary classifier that distinguishes real data from generated ones, and to update the generator ($G_\theta(\boldsymbol{z})$) to fool it.
- The discriminator is trained to classify data from the following two classes:

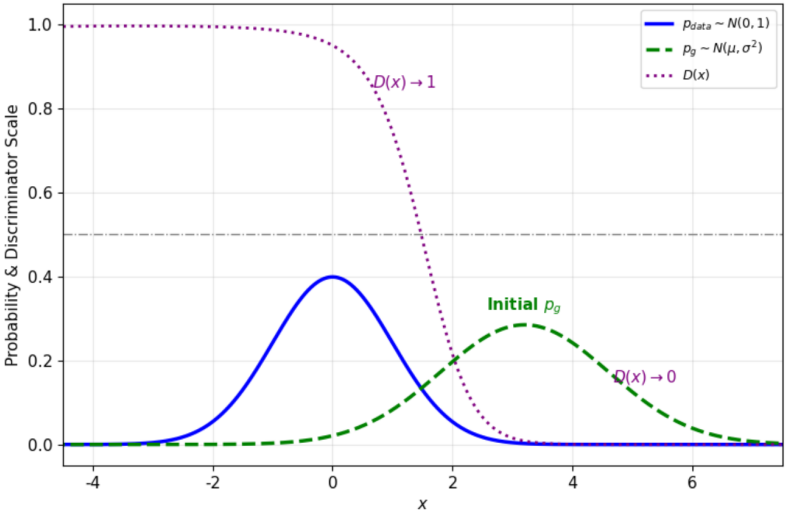
$$(\text{Real Data}; Y = 1)\ \boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \sim p(\boldsymbol{x}) \tag{3}$$

and

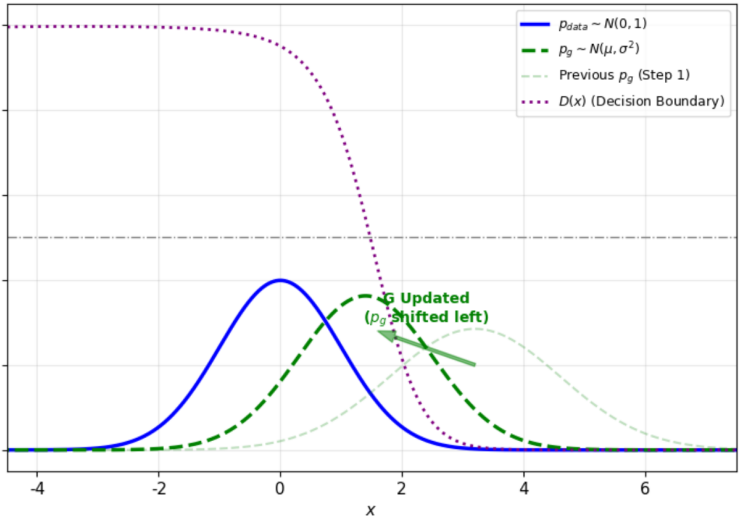
$$(\text{Generated Data}; Y = 0)\ G_\theta(\boldsymbol{z}_1), \ldots, G_\theta(\boldsymbol{z}_n),\ \text{where}\ \boldsymbol{z}_1, \ldots, \boldsymbol{z}_n \sim p(\boldsymbol{z}). \tag{4}$$

- We denote the output of the classifier (or *discriminator*) as $D_\phi(\boldsymbol{x})$, which models $P(Y = 1 | \boldsymbol{X} = \boldsymbol{x})$.
- The generator G_θ is then updated to maximize the loss of the discriminator D_ϕ. This process is iteratively repeated, which is referred to as *adversarial learning*.

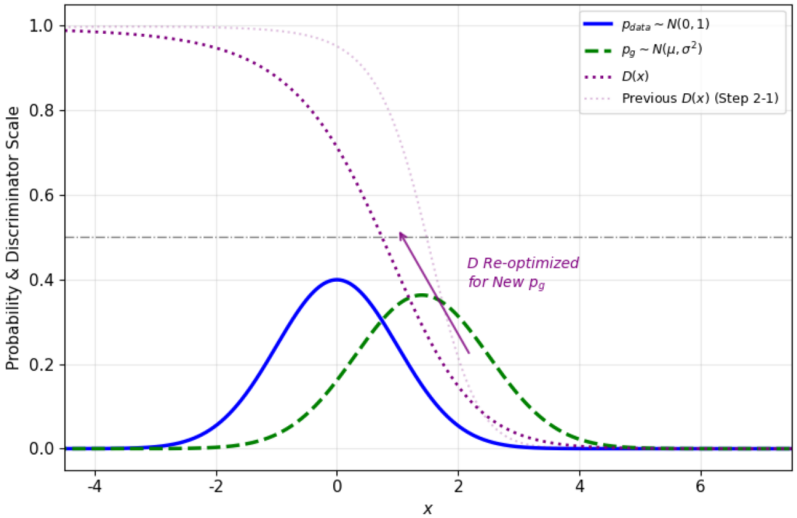
Adversarial Learning



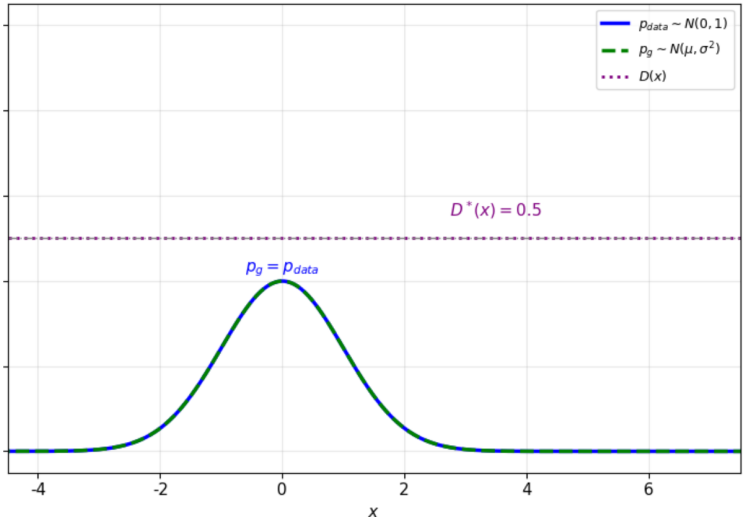
Adversarial Learning



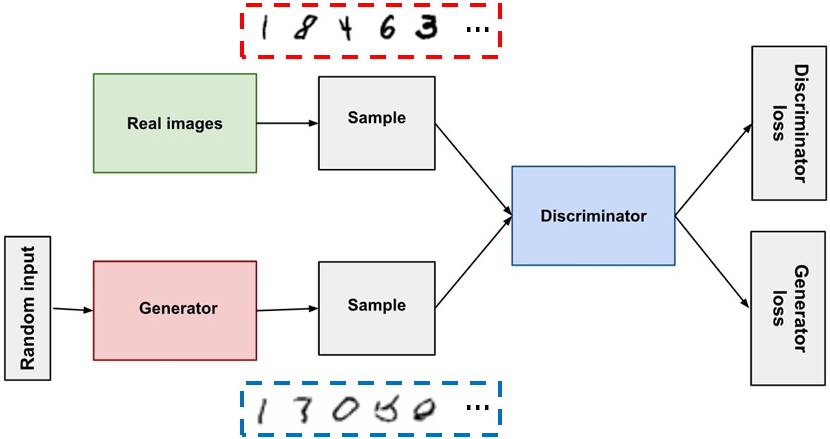
Adversarial Learning



Adversarial Learning



Adversarial Learning



Images were edited from
https://developers.google.com/machine-learning/gan/gan_structure and
https://github.com/MorvanZhou/mnistGANs.

Adversarial Learning

- Adversarial learning can be formulated as follows.
- First, the loss function of D_ϕ is the negative log-likelihood (or *cross-entropy loss*):

$$l(D_\phi(\boldsymbol{x}), y) = -I(y = 1) \log D_\phi(\boldsymbol{x}) - I(y = 0) \log \big(1 - D_\phi(\boldsymbol{x})\big). \tag{5}$$

- With the dataset consisting of n real data and n generated data, $(\boldsymbol{x}_1, y_1 = 1), \ldots, (\boldsymbol{x}_n, y_n = 1), (G_\theta(\boldsymbol{z}_1), y_{n+1} = 0), \ldots, (G_\theta(\boldsymbol{z}_n), y_{2n} = 0)$, the average loss (or *empirical risk*) can be expressed as:

$$-n^{-1} \sum_{i=1}^{n} \log D_\phi(\boldsymbol{x}_i) - n^{-1} \sum_{i=1}^{n} \log(1 - D_\phi(G_\theta(\boldsymbol{z}_i))). \tag{6}$$

- Next, the loss function of the generator G_θ is $-l(D_\phi(G_\theta(\boldsymbol{z})), y = 0)$, and the objective function can be expressed as:

$$n^{-1} \sum_{i=1}^{n} \log(1 - D_\phi(G_\theta(\boldsymbol{z}_i))). \tag{7}$$

Adversarial Learning

Algorithm 1 Minibatch stochastic gradient descent training of generative adversarial nets. The number of steps to apply to the discriminator, k, is a hyperparameter. We used $k = 1$, the least expensive option, in our experiments.

for number of training iterations **do**

  for k steps **do**

  - Sample minibatch of m noise samples $\{\boldsymbol{z}^{(1)}, \ldots, \boldsymbol{z}^{(m)}\}$ from noise prior $p_g(\boldsymbol{z})$.
  - Sample minibatch of m examples $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}$ from data generating distribution $p_{\text{data}}(\boldsymbol{x})$.
  - Update the discriminator by ascending its stochastic gradient:

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^{m} \left[\log D\left(\boldsymbol{x}^{(i)}\right) + \log\left(1 - D\left(G\left(\boldsymbol{z}^{(i)}\right)\right)\right) \right].$$

  end for

  - Sample minibatch of m noise samples $\{\boldsymbol{z}^{(1)}, \ldots, \boldsymbol{z}^{(m)}\}$ from noise prior $p_g(\boldsymbol{z})$.
  - Update the generator by descending its stochastic gradient:

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^{m} \log\left(1 - D\left(G\left(\boldsymbol{z}^{(i)}\right)\right)\right).$$

end for

The gradient-based updates can use any standard gradient-based learning rule. We used momentum in our experiments.

The pseudo-algorithm is from Goodfellow et al. (2014).

Adversarial Learning

- The adversarial learning process involves alternately maximizing and minimizing the negative cross-entropy loss:

$$V(\theta, \phi) := \int_{\mathbb{R}^d} \big(\log D_\phi(\boldsymbol{x}) \big) p(\boldsymbol{x})\, d\boldsymbol{x} + \int_{\mathbb{R}^d} \big(\log(1 - D_\phi(\boldsymbol{x})) \big) p_\theta(\boldsymbol{x})\, d\boldsymbol{x}. \tag{8}$$

- This process can be viewed as a two-player minimax game where the goal is to find

$$\theta^* \in \arg\min_\theta \big(\max_\phi V(\theta, \phi) \big). \tag{9}$$

- The adversarial learning can be interpreted as repeated cycles of approximating and minimizing the Jensen-Shannon (JS) divergence:

$$\max_\phi V(\theta, \phi) = \mathcal{D}_{\mathsf{JS}}(p \parallel p_\theta) := \frac{1}{2} \left(\mathcal{D}_{\mathsf{KL}} \left(p || \frac{p + p_\theta}{2} \right) + \mathcal{D}_{\mathsf{KL}} \left(p_\theta || \frac{p + p_\theta}{2} \right) \right) \tag{10}$$

up to a constant addition and sign-preserving multiplication.

Adversarial Learning

Exercise 1

Let the adversarial objective be:

$$V(\theta, \phi) := \int_{\mathbb{R}^d} \big(\log D_\phi(\boldsymbol{x})\big) p(\boldsymbol{x})\, d\boldsymbol{x} + \int_{\mathbb{R}^d} \big(\log(1 - D_\phi(\boldsymbol{x}))\big) p_\theta(\boldsymbol{x})\, d\boldsymbol{x}. \quad (11)$$

Assume the discriminator family $\{D_\phi \mid \phi \in \Phi\}$ is sufficiently expressive such that for any θ, there exists $\phi^*(\theta) \in \Phi$ satisfying:

$$D_{\phi^*(\theta)}(\boldsymbol{x}) = \frac{p(\boldsymbol{x})}{p(\boldsymbol{x}) + p_\theta(\boldsymbol{x})}. \quad (12)$$

Adversarial Learning

1. Show that, for any $a \in (0,1)$, $b \in \mathbb{R}^+$, and $c \in \mathbb{R}^+$,

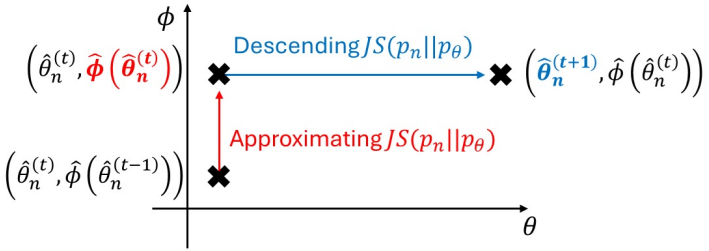
$$b \log a + c \log(1-a) \leq b \log \frac{b}{b+c} + c \log \frac{c}{b+c}, \tag{13}$$

where equality holds if and only if $a = b/(b+c)$.

Adversarial Learning

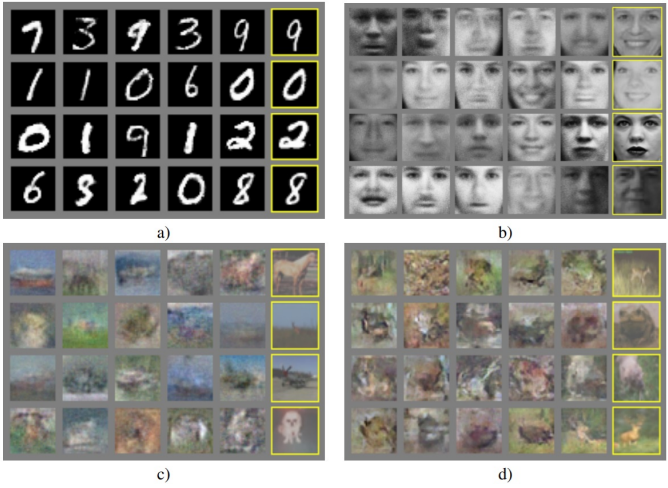
2. Show that $\max_\phi V(\theta, \phi) = 2\mathcal{D}_{\mathsf{JS}}(p \parallel p_\theta) - \log 4$.

Adversarial Learning



- The two adversarial networks, the discriminator and the generator, are trained alternately:
  1. Given $\hat{\theta}_n^{(t)}$, update the discriminator to obtain $\hat{\phi}(\hat{\theta}_n^{(t)})$ by maximizing $V(\hat{\theta}_n^{(t)}, \phi)$.
  2. Given $\hat{\phi}(\hat{\theta}_n^{(t)})$, update the generator to obtain $\hat{\theta}_n^{(t+1)}$ by minimizing $V(\theta, \hat{\phi}(\hat{\theta}_n^{(t)}))$.
  3. Repeat the above processes.

Generation Result



Images are from Goodfellow et al. (2014).

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Representation Learning with Adversarial Learning

- In (vanilla) GANs, there is no density-based inference, as they do not have explicit density models.
- Several variants of GANs have been proposed to learn (interpretable) representations and conduct corresponding inferences.

Information Maximizing Generative Adversarial Network

- Information Maximizing GANs (InfoGANs) (Chen et al., 2016) decompose the latent factors into two parts: (i) signal-related factors ($\boldsymbol{C}$) and (ii) noise-related factors ($\boldsymbol{Z}$).
- They modify the generator to $G_\theta(\boldsymbol{Z}, \boldsymbol{C})$ and the value function as:

$$V(\theta, \phi) - \lambda \, \mathrm{MI}(\boldsymbol{C}; G_\theta(\boldsymbol{Z}, \boldsymbol{C})) \tag{14}$$

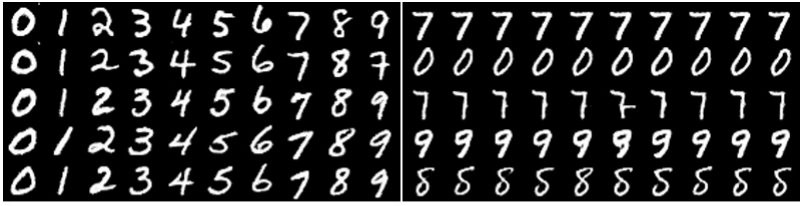
where $\mathrm{MI}(\boldsymbol{C}; G_\theta(\boldsymbol{Z}, \boldsymbol{C})) := \mathcal{D}_{\mathrm{KL}}(p(\boldsymbol{c})p_\theta(G_\theta(\boldsymbol{z}, \boldsymbol{c})|\boldsymbol{c})||p(\boldsymbol{c})p(G_\theta(\boldsymbol{z}, \boldsymbol{c})))$ represents the mutual information between the signal-related factor $\boldsymbol{C}$ and the generated data $G_\theta(\boldsymbol{Z}, \boldsymbol{C})$, and λ is a hyperparameter.

- The new penalty term is constant w.r.t. ϕ, so the maximization over ϕ yields:

$$\big(2\mathcal{D}_{\mathrm{JS}}(p||p_\theta) - \log 4\big) - \lambda \, \mathrm{MI}(\boldsymbol{C}; G_\theta(\boldsymbol{Z}, \boldsymbol{C})). \tag{15}$$

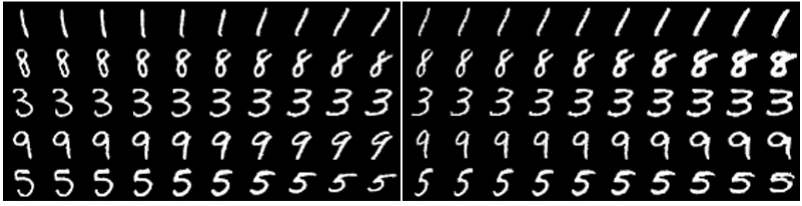
The penalty encourages $\boldsymbol{C}$ to be dependent on $G_\theta(\boldsymbol{Z}, \boldsymbol{C})$.

InfoGAN: Generation Result



(a) Varying c_1 on InfoGAN (Digit type)

(b) Varying c_1 on regular GAN (No clear meaning)



(c) Varying c_2 from -2 to 2 on InfoGAN (Rotation)

(d) Varying c_3 from -2 to 2 on InfoGAN (Width)

Images are from Chen et al. (2016).

Bidirectional Generative Adversarial Network

- Bidirectional GANs (BiGANs) (Donahue et al., 2016) introduce an encoder $E_\psi(\boldsymbol{x})$ to directly learn the dimension-reduction mapping.[1]
- The discriminator is modified to classify pairs $(\boldsymbol{X}, E_\psi(\boldsymbol{X}))$ vs. $(G_\theta(\boldsymbol{Z}), \boldsymbol{Z})$. The corresponding value function can be expressed as:

$$V(\theta, \phi, \psi) := \int_{\mathbb{R}^d} \big(\log D_\phi(\boldsymbol{x}, E_\psi(\boldsymbol{x}))\big) p(\boldsymbol{x})\, d\boldsymbol{x} + \int_{\mathbb{R}^r} \big(\log(1 - D_\phi(G_\theta(\boldsymbol{z}), \boldsymbol{z}))\big) p(\boldsymbol{z})\, d\boldsymbol{z}. \tag{16}$$

[1] Adversarially Learned Inference (Dumoulin et al., 2016) is a concurrently proposed work with a similar idea.

Bidirectional Generative Adversarial Network

Exercise 2

Let $p_\psi(\boldsymbol{x}, \boldsymbol{z}) := p(\boldsymbol{x})p_\psi(\boldsymbol{z} \mid \boldsymbol{x})$ denote the joint distribution induced by the encoder, and let $p_\theta(\boldsymbol{x}, \boldsymbol{z}) := p(\boldsymbol{z})p_\theta(\boldsymbol{x} \mid \boldsymbol{z})$ denote the joint distribution induced by the generator. The BiGAN minimax objective is given by:

$$V(\theta, \phi, \psi) := \mathrm{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p_\psi}\left[\log D_\phi(\boldsymbol{x}, \boldsymbol{z})\right] + \mathrm{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p_\theta}\left[\log(1 - D_\phi(\boldsymbol{x}, \boldsymbol{z}))\right].$$

Assume the discriminator class $\{D_\phi\}$ is sufficiently expressive such that for any (θ, ψ), there exists $\phi^*(\theta, \psi)$ satisfying:

$$D_{\phi^*(\theta,\psi)}(\boldsymbol{x}, \boldsymbol{z}) = \frac{p_\psi(\boldsymbol{x}, \boldsymbol{z})}{p_\psi(\boldsymbol{x}, \boldsymbol{z}) + p_\theta(\boldsymbol{x}, \boldsymbol{z})}. \tag{17}$$

Bidirectional Generative Adversarial Network

Show that $\max_\phi V(\theta, \phi, \psi) = 2\mathcal{D}_{\mathsf{JS}}(p_\psi \parallel p_\theta) - \log 4$.

Bidirectional Generative Adversarial Network

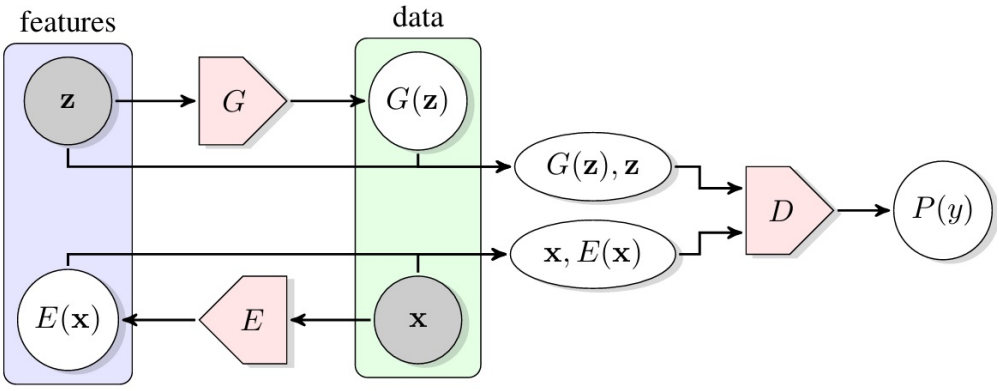
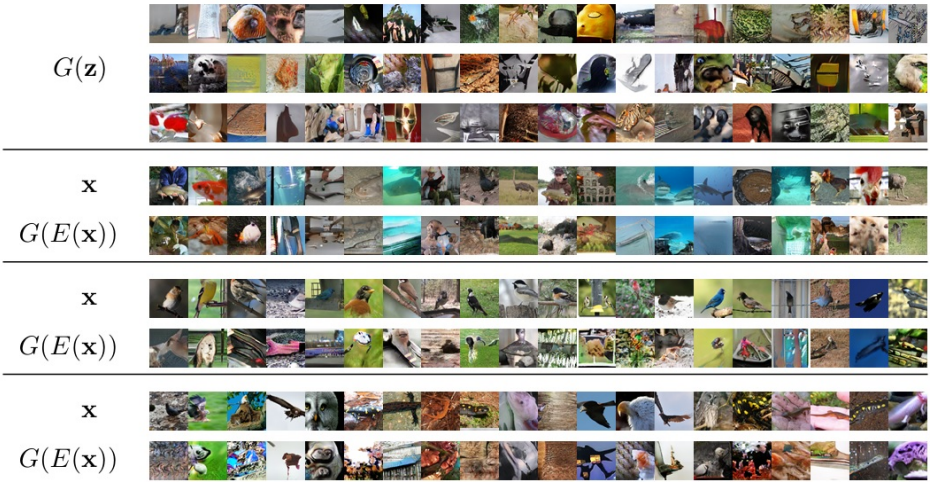


Figure 1: The structure of Bidirectional Generative Adversarial Networks (BiGAN).

Images are from Donahue et al. (2016).

Bidirectional Generative Adversarial Network



Images are from Donahue et al. (2016).

Bidirectional Generative Adversarial Network

- The maximized value function over ϕ is $\mathcal{D}_{\mathrm{JS}}(p(\boldsymbol{x}, E_\psi(\boldsymbol{x}))||p_\theta(G_\theta(\boldsymbol{z}), \boldsymbol{z}))$, and is minimized w.r.t. encoder and generator parameters (θ, ψ).
- It is known that the optimal encoder and generator identify an ideal autoencoder that perfectly reconstruct $\boldsymbol{x}$.

Adversarially Learned Anomaly Detection

- Despite the optimal encoder and generator perfectly reconstruct $\boldsymbol{x}$, it has been observed that this may empirically not hold (Li et al., 2017).
- Adversarially Learned Anomaly Detection (Zenati et al., 2018) introduces adversarial learning about the reconstructed results:

$$
\begin{aligned}
&V(\theta, \phi, \psi) \\
&:= \int_{\mathbb{R}^d} \big(\log D_{XZ,\phi}(\boldsymbol{x}, E_\psi(\boldsymbol{x})) \big) p(\boldsymbol{x})\, d\boldsymbol{x} + \int_{\mathbb{R}^r} \big(\log(1 - D_{XZ,\phi}(G_\theta(\boldsymbol{z}), \boldsymbol{z})) \big) p(\boldsymbol{z})\, d\boldsymbol{z} \\
&\quad + \int_{\mathbb{R}^d} \big(\log D_{XX,\phi}(\boldsymbol{x}, \boldsymbol{x}) \big) p(\boldsymbol{x})\, d\boldsymbol{x} + \int_{\mathbb{R}^d} \big(\log(1 - D_{XX,\phi}(\boldsymbol{x}, G_\theta(E_\psi(\boldsymbol{x})))) \big) p(\boldsymbol{x})\, d\boldsymbol{x} \\
&\quad + \int_{\mathbb{R}^r} \big(\log D_{ZZ,\phi}(\boldsymbol{z}, \boldsymbol{z}) \big) p(\boldsymbol{z})\, d\boldsymbol{z} + \int_{\mathbb{R}^r} \big(\log(1 - D_{ZZ,\phi}(\boldsymbol{z}, E_\psi(G_\theta(\boldsymbol{z})))) \big) p(\boldsymbol{z})\, d\boldsymbol{z}.
\end{aligned}
\tag{18}
$$

Here, $D_{XZ,\phi}$, $D_{XX,\phi}$, and $D_{ZZ,\phi}$ are all discriminators.

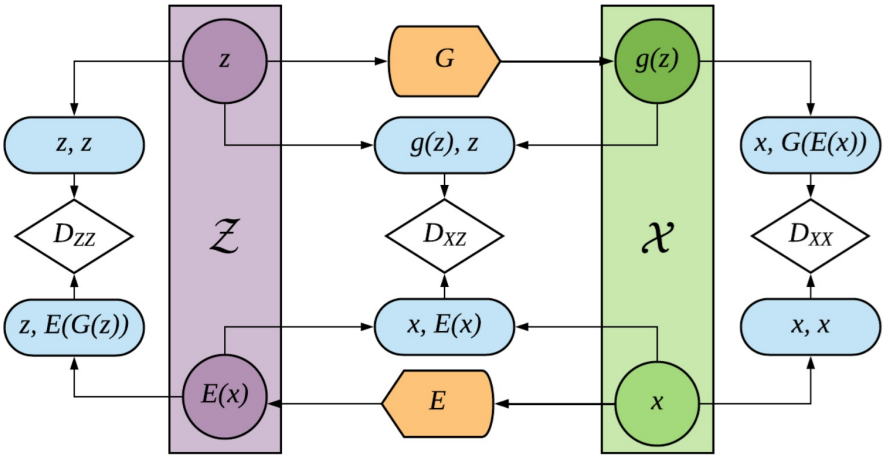
Adversarially Learned Anomaly Detection

- The maximized value function over the discriminator parameter ϕ is:

$$\begin{aligned}\max_{\phi} V(\theta, \phi, \psi) = &\big(2\mathcal{D}_{\mathrm{JS}}(p(\boldsymbol{x}, E_\psi(\boldsymbol{x}))||p_\theta(G_\theta(\boldsymbol{z}), \boldsymbol{z})) - \log 4\big) \\ &+ \big(2\mathcal{D}_{\mathrm{JS}}(p(\boldsymbol{x}, \boldsymbol{x})||p_{\theta,\psi}(\boldsymbol{x}, G_\theta(E_\psi(\boldsymbol{x})))) - \log 4\big) \\ &+ \big(2\mathcal{D}_{\mathrm{JS}}(p(\boldsymbol{z}, \boldsymbol{z})||p_{\theta,\psi}(\boldsymbol{z}, E_\psi(G_\theta(\boldsymbol{z})))) - \log 4\big).\end{aligned} \tag{19}$$

- The optimal encoder and generator still identify an ideal autoencoder.

Adversarially Learned Anomaly Detection



Images are from Zenati et al. (2018).

Adversarially Learned Anomaly Detection

Algorithm 1 Adversarially Learned Anomaly Detection

Input $\boldsymbol{x}, \sim p_{\mathcal{X}_{Test}}(x), E, G, f_{xx}$ where f_{xx} is the feature layer of D_{xx}

Output $A(\boldsymbol{x})$, where A is the anomaly score

1: **procedure** INFERENCE
2: $\tilde{\boldsymbol{z}} \leftarrow E(\boldsymbol{x})$ ▷ Encode samples
3: $\hat{\boldsymbol{x}} \leftarrow G(\tilde{\boldsymbol{z}})$, ▷ Reconstruct samples
4: $f_\delta \leftarrow f_{xx}(\boldsymbol{x}, \hat{\boldsymbol{x}})$
5: $f_\alpha \leftarrow f_{xx}(\boldsymbol{x}, \boldsymbol{x})$
6: return $\|f_\delta - f_\alpha\|_1$
7: **end procedure**

The algorithm is from Zenati et al. (2018).

Adversarially Learned Anomaly Detection



Figure 5. Error analysis on SVHN when the normal class is the "3" digit. 1st row: normal data; 2nd row: reconstruction of normal data; 3rd row: anomalous data; 4th row: reconstruction of anomalous data.

Images are from Zenati et al. (2018).

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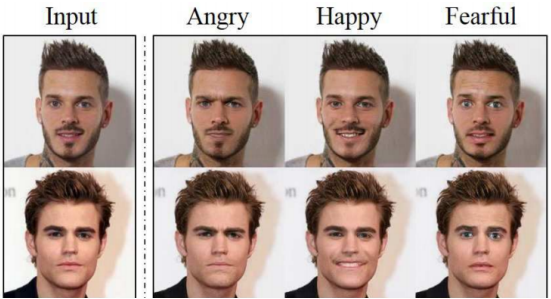
4. Other Advanced Topics

Data-to-Data Translation

Conditioning Data ($\boldsymbol{C}$)	Target Data ($\boldsymbol{X}$)	Name of Corresponding Task
Image	Image	Image-to-Image translation
Video	Video	Video-to-Video translation
Text	Text	Text-to-Text translation
Text	Image	Text-to-Image translation

- Data-to-Data translation refers to synthesizing one type of data (from the *source domain*) using another (from the *target domain*).

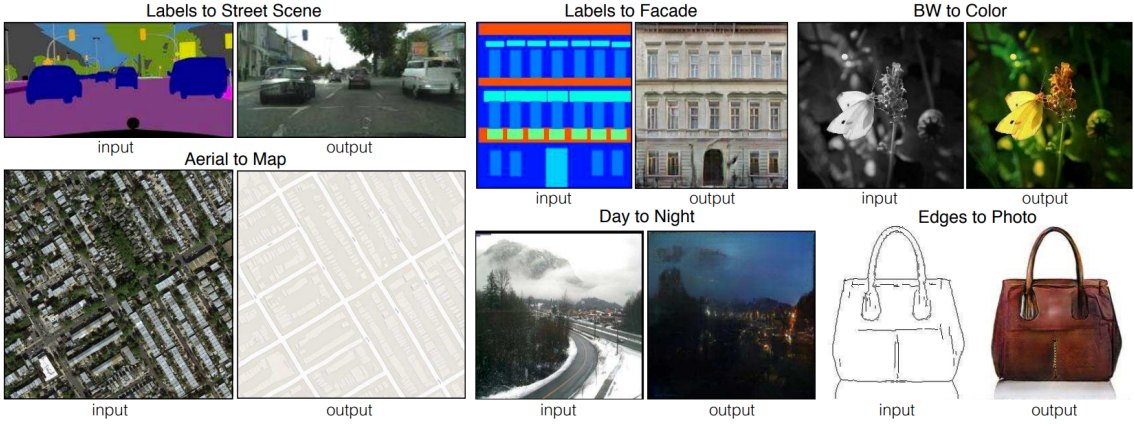
Data-to-Data Translation





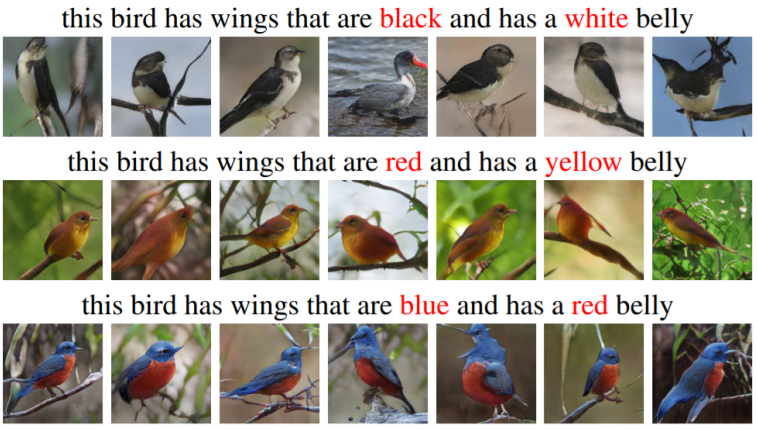
Images are edited from Choi et al. (2018) (left) and Choi et al. (2020) (right).

Data-to-Data Translation



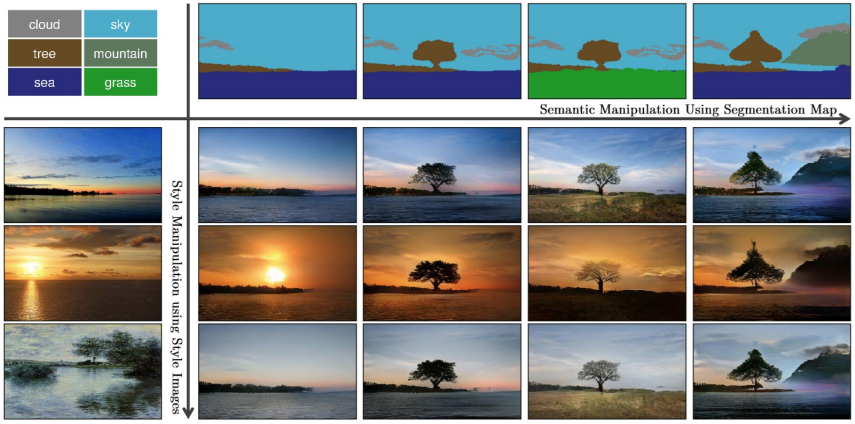
Summary of applications of Image-to-Image translation (Isola et al., 2017).

Data-to-Data Translation



Images are from Xu et al. (2018).

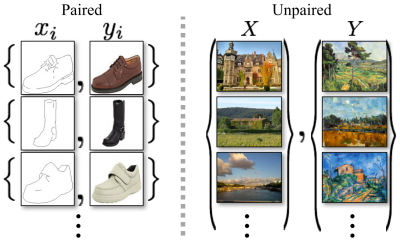
Data-to-Data Translation



- Data-to-Data Translation is an example of *Controllable Generation*. Changes in the conditioning data influence the specific characteristics of the generated results.

Images are from Park et al. (2019).

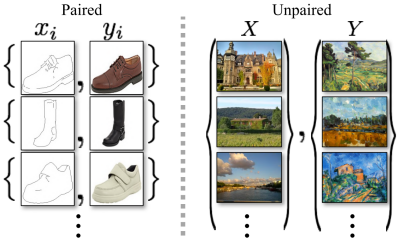
Paired vs. Unpaired Datasets



- According to the presence of pair information, methods are divided into unpaired and paired settings.
- A dataset is *paired* if both source and target data (modalities) are observed for every subject, and pair information (ID of subjects) is provided. If the pair information is not provided, the dataset is *unpaired*.

The figure is from Zhu et al. (2017).

Paired vs. Unpaired Datasets



- The pair of a given source datum usually indicates the most similar datum in the target dataset. The criterion of 'similar' depends on the context.
- Especially for unpaired datasets, it is difficult to define pairs. For example, when the source and target data are horse and zebra images, respectively, no object possesses the phenotype of both a horse and a zebra simultaneously.

The figure is from Zhu et al. (2017).

Paired vs. Unpaired Datasets

- Earliest works usually assume a one-to-one relationship between source and target data, regardless of whether the training dataset contains the pair information.
- The translator is referred to as a "generator" and denoted by $G_\theta(\boldsymbol{C})$ since it learns the data distribution ($p(\boldsymbol{x})$), but its role is more of a transformer than a generator.
- However, the mapping from source data to target data is often one-to-many. For example, when we translate daytime scenes to their nighttime counterparts, there can be multiple nighttime scenes for a given daytime scene.
- Recent works have introduced latent factors $\boldsymbol{Z}$ to extend the translator $G_\theta(\boldsymbol{C})$ to $G_\theta(\boldsymbol{Z}, \boldsymbol{C})$. This research field is often called *multimodal synthesis*.

Paired Data-to-Data Translation

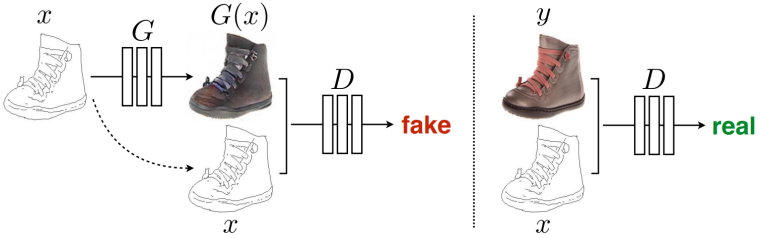


Figure: The architecture of Pix2Pix (Isola et al., 2017).

- We denote paired datasets by $\{(\boldsymbol{x}_i, \boldsymbol{c}_i)\}_{i=1}^n$. The generator G_θ inputs $\boldsymbol{c}_i$ to synthesize datum similar to $\boldsymbol{x}_i$.
- This can be viewed as an multivariate regression utilizing adversarial learning.

Paired Data-to-Data Translation

- The goal is to learn a translator G_θ s.t. $G_\theta(\boldsymbol{C}) = \boldsymbol{X}$. For example, in Pix2Pix (Isola et al., 2017), the value function is:[2]

$$V(\theta, \phi) := \int_{\mathbb{R}^{d_C + d_X}} \left(\log D_\phi(\boldsymbol{c}, \boldsymbol{x})\right) p(\boldsymbol{c}, \boldsymbol{x}) d\boldsymbol{c} d\boldsymbol{x} + \int_{\mathbb{R}^{d_C}} \left(\log\left(1 - D_\phi(\boldsymbol{c}, G_\theta(\boldsymbol{c}))\right)\right) p(\boldsymbol{c}) d\boldsymbol{c}. \tag{20}$$

- The objective function is:

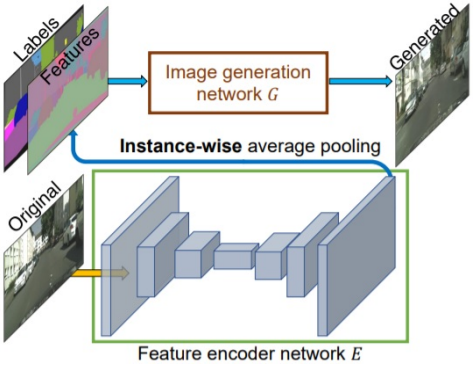
$$\int_{\mathbb{R}^{d_C + d_X}} \|\boldsymbol{x} - G_\theta(\boldsymbol{c})\|_1 p(\boldsymbol{c}, \boldsymbol{x}) d\boldsymbol{c} d\boldsymbol{x} + \lambda V(\theta, \phi) \tag{21}$$

  where the first term is the prediction error and λ is a hyperparameter.

- Since the objective includes $p(\boldsymbol{c}, \boldsymbol{x})$, we need paired data to use it.

[2] The authors applied dropout, a stochastic layer, to build a probabilistic G_θ. For simplicity, we omit this stochastic part in the formulation in this lecture.

Paired Data-to-Data Translation



- Pix2PixHD (Wang et al., 2018) is an extension of Pix2Pix and introduces a feature encoder network to address multi-modal synthesis problem.

The figure is from Wang et al. (2018).

Paired Data-to-Data Translation

- The value function is:

$$V(\theta, \phi, \psi) := \int_{\mathbb{R}^{d_C + d_X}} \left(\log D_\phi(\boldsymbol{c}, \boldsymbol{x})\right) p(\boldsymbol{c}, \boldsymbol{x}) d\boldsymbol{c} d\boldsymbol{x} + \int_{\mathbb{R}^{C+\mathbb{X}}} \left(\log\left(1 - D_\phi(\boldsymbol{c}, G_\theta(\boldsymbol{c}, E_\psi(\boldsymbol{x})))\right)\right) p(\boldsymbol{c}, \boldsymbol{x}) d\boldsymbol{c} d\boldsymbol{x}. \tag{22}$$

- Pix2PixHD introduces latent variables, but their distribution is not estimated.
- The K-means clustering algorithm is applied to latent variables extracted from the training dataset. At inference time, one of K centroids is randomly sampled to generate target data.

Paired Data-to-Data Translation

Exercise 3

Let $p(\boldsymbol{c}, \boldsymbol{x}) := p(\boldsymbol{c})p(\boldsymbol{x} \mid \boldsymbol{c})$ denote the joint distribution of condition and data, and let $p_{\theta,\psi}(\boldsymbol{c}, \boldsymbol{x}) := p(\boldsymbol{c})p_{\theta,\psi}(\boldsymbol{x} \mid \boldsymbol{c})$ denote the joint distribution induced by the generator. The conditional minimax objective is given by:

$$V(\theta, \phi, \psi) := \mathrm{E}_{(\boldsymbol{c},\boldsymbol{x})\sim p}\left[\log D_\phi(\boldsymbol{c}, \boldsymbol{x})\right] + \mathrm{E}_{(\boldsymbol{c},\boldsymbol{x})\sim p_{\theta,\psi}}\left[\log(1 - D_\phi(\boldsymbol{c}, \boldsymbol{x}))\right].$$

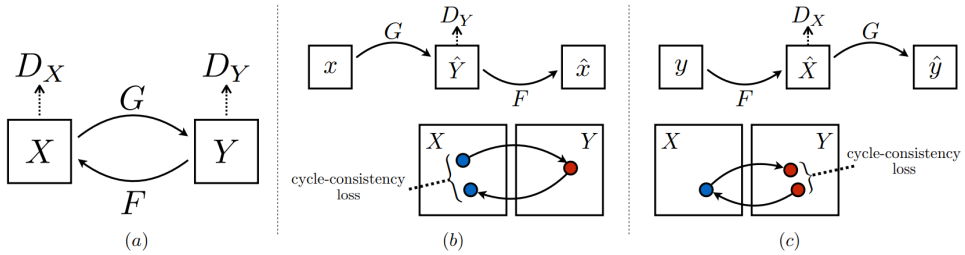
Assume the discriminator class $\{D_\phi\}$ is sufficiently expressive such that for any (θ, ψ), there exists $\phi^*(\theta, \psi)$ satisfying:

$$D_{\phi^*(\theta,\psi)}(\boldsymbol{c}, \boldsymbol{x}) = \frac{p(\boldsymbol{c}, \boldsymbol{x})}{p(\boldsymbol{c}, \boldsymbol{x}) + p_{\theta,\psi}(\boldsymbol{c}, \boldsymbol{x})}. \tag{23}$$

Paired Data-to-Data Translation

Show that $\max_\phi V(\theta, \phi, \psi) = 2\mathcal{D}_{\mathsf{JS}}(p \parallel p_{\theta,\psi}) - \log 4$.

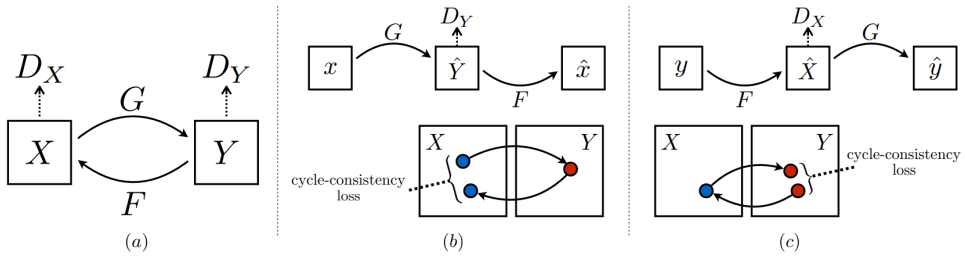
Unpaired Data-to-Data Translation



- The pair information is usually not provided.
- CycleGAN (Zhu et al., 2017) is one of the earliest unpaired data-to-data translation models. It trains two translators and utilizes the cycle-consistency loss.

The figure is from Zhu et al. (2017).

Unpaired Data-to-Data Translation



- CycleGANs have a pair of translators: one from the source domain to the target ($G_{C\to X,\theta}(\boldsymbol{c})$) and the other from the target back to the source ($G_{X\to C,\theta}(\boldsymbol{x})$).
- The goal is to find $(G_{C\to X,\theta}, G_{X\to C,\theta})$:
  1. (Transportation) $p_\theta(G_{C\to X,\theta}(\boldsymbol{c})) = p(\boldsymbol{x})$ and $p_\theta(G_{X\to C,\theta}(\boldsymbol{x})) = p(\boldsymbol{c})$.
  2. (Cycle-Consistency) $(G_{C\to X,\theta} \circ G_{X\to C,\theta})(\boldsymbol{c}) = \boldsymbol{c}$ and $(G_{X\to C,\theta} \circ G_{C\to X,\theta})(\boldsymbol{x}) = \boldsymbol{x}$.

The figure is from Zhu et al. (2017).

Unpaired Data-to-Data Translation

- The value function can be expressed as:

$$V(\theta,\phi) := \int_{\mathbb{R}^{d_X}} \left(\log D_{X,\phi}(\boldsymbol{x})\right) p(\boldsymbol{x}) d\boldsymbol{x} + \int_{\mathbb{R}^{d_C}} \left(\log\left(1 - D_{X,\phi}(G_{C\to X,\theta}(\boldsymbol{c}))\right)\right) p(\boldsymbol{c}) d\boldsymbol{c} + \int_{\mathbb{R}^{d_C}} \left(\log D_{C,\phi}(\boldsymbol{c})\right) p(\boldsymbol{c}) d\boldsymbol{c} + \int_{\mathbb{R}^{d_X}} \left(\log\left(1 - D_{C,\phi}(G_{X\to C,\theta}(\boldsymbol{x}))\right)\right) p(\boldsymbol{x}) d\boldsymbol{x}.$$

- They introduce cycle-consistency loss:

$$\int_{\mathbb{R}^{d_C}} \|(G_{C\to X,\theta}\circ G_{X\to C,\theta})(\boldsymbol{c}) - \boldsymbol{c}\|_1 p(\boldsymbol{c}) d\boldsymbol{c} + \int_{\mathbb{R}^{d_X}} \|(G_{X\to C,\theta}\circ G_{C\to X,\theta})(\boldsymbol{x}) - \boldsymbol{x}\|_1 p(\boldsymbol{x}) d\boldsymbol{x}.$$

  The cycle-consistency loss encourages the inverse relation between two translators.

- The loss function also includes the identity mapping loss term:

$$\int_{\mathbb{R}^{d_X}} \|\boldsymbol{x} - G_{X\to C,\theta}(\boldsymbol{x})\| p(\boldsymbol{x}) d\boldsymbol{x} + \int_{\mathbb{R}^{d_C}} \|\boldsymbol{c} - G_{C\to X,\theta}(\boldsymbol{c})\| p(\boldsymbol{c}) d\boldsymbol{c}.$$

  This approach can be viewed as learning pair information by neural networks.

Unpaired Data-to-Data Translation

Exercise 4

Let $p_X(\boldsymbol{x})$ and $p_C(\boldsymbol{c})$ denote the data distributions in domain $\mathcal{X}$ and domain $\mathcal{C}$, respectively. Let $p_{X,\theta}(\boldsymbol{x})$ and $p_{C,\theta}(\boldsymbol{c})$ denote the marginal distributions induced by the generators $G_{C\to X,\theta}$ and $G_{X\to C,\theta}$. The adversarial objective in CycleGAN is defined as:

$$\begin{aligned} V(\theta,\phi) := \mathrm{E}_{\boldsymbol{x}\sim p_X}\left[\log D_{X,\phi}(\boldsymbol{x})\right] + \mathrm{E}_{\boldsymbol{x}\sim p_{X,\theta}}\left[\log(1-D_{X,\phi}(\boldsymbol{x}))\right] \\ + \mathrm{E}_{\boldsymbol{c}\sim p_C}\left[\log D_{C,\phi}(\boldsymbol{c})\right] + \mathrm{E}_{\boldsymbol{c}\sim p_{C,\theta}}\left[\log(1-D_{C,\phi}(\boldsymbol{c}))\right]. \end{aligned}$$

Assume the discriminator class $\{D_{X,\phi}, D_{C,\phi}\}$ is sufficiently expressive such that for any θ, there exists $\phi^*(\theta)$ satisfying:

$$D_{X,\phi^*(\theta)}(\boldsymbol{x}) = \frac{p_X(\boldsymbol{x})}{p_X(\boldsymbol{x}) + p_{X,\theta}(\boldsymbol{x})}, \quad D_{C,\phi^*(\theta)}(\boldsymbol{c}) = \frac{p_C(\boldsymbol{c})}{p_C(\boldsymbol{c}) + p_{C,\theta}(\boldsymbol{c})}. \tag{24}$$

Unpaired Data-to-Data Translation

Show that the optimal discriminator objective achieves:

$$\max_{\phi} V(\theta, \phi) = 2\mathcal{D}_{\mathrm{JS}}(p_X \parallel p_{X,\theta}) + 2\mathcal{D}_{\mathrm{JS}}(p_C \parallel p_{C,\theta}) - \log 16. \tag{25}$$

Unpaired Data-to-Data Translation

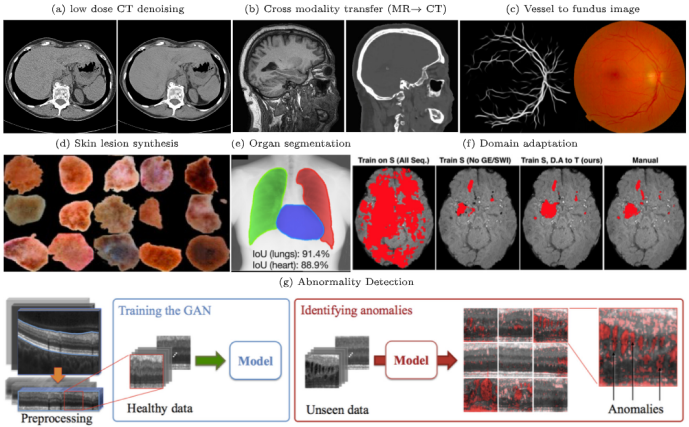


Figure: Important applications include image reconstruction, segmentation, detection, classification, and cross-modality synthesis (Yi et al., 2019)

Unpaired Data-to-Data Translation: Low Dose CT Denoising

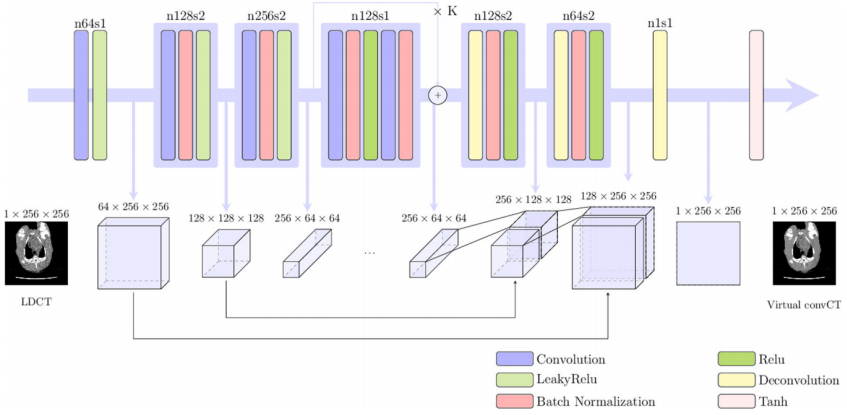


Figure: Low-dose CTs are feed-forwarded by a U-net to synthesize conventional CTs (Yi and Babyn, 2018). The dataset is a paired dataset.

Unpaired Data-to-Data Translation: Cross modality transfer (MR → CT)

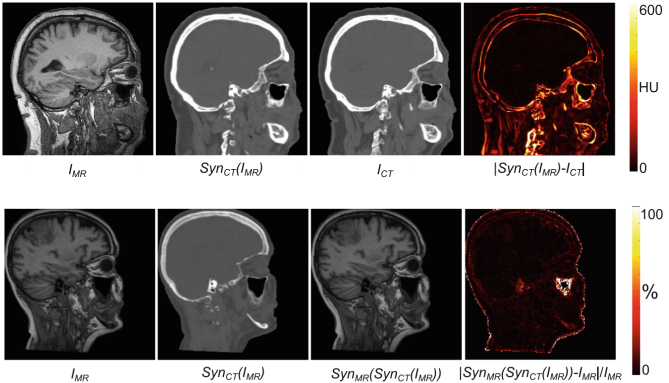


Figure: Wolterink et al. (2017) trained CycleGAN to translate MR images into CT images. The translation from CT to MR ($(2,2) \rightarrow (2,3)$) is also successful.

Unpaired Data-to-Data Translation: Cross modality transfer (Between MRIs)

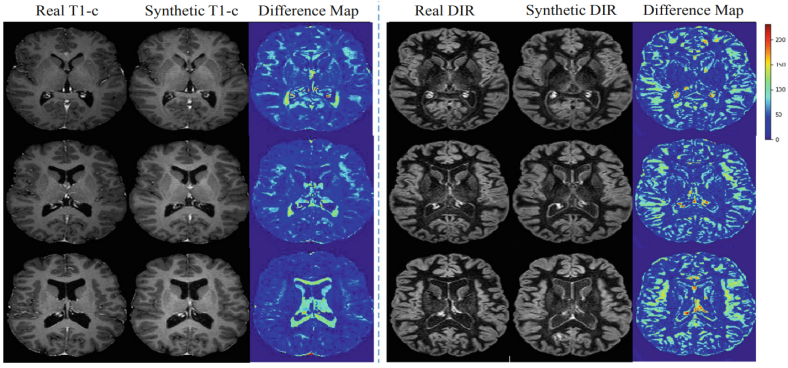


Figure: Li et al. (2019) trained a multi-CycleGAN between various modalities to translate MR images into other modalities. Left: $\{\text{T1}, \text{T2}, \text{Flair}\} \rightarrow$ Contrast-enhanced T1; Right: $\{\text{T1}, \text{T2}, \text{Flair}\} \rightarrow$ Double inversion recovery

Unpaired Data-to-Data Translation: Organ Segmentation

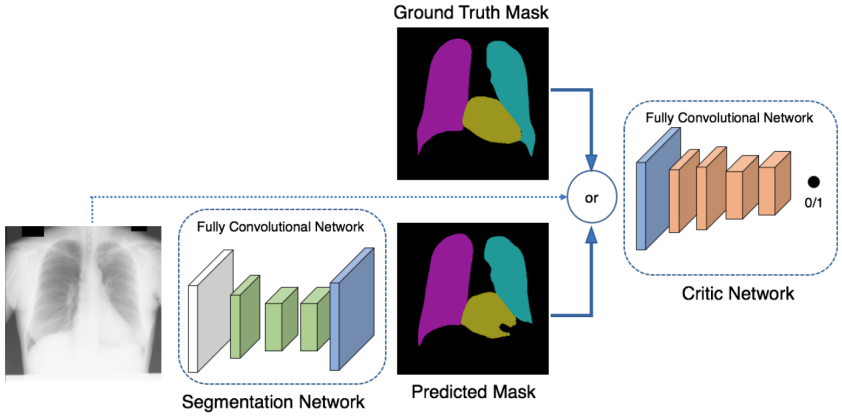


Figure: Dai et al. (2018) viewed the segmentation model as a conditional image synthesis model and used the summation of segmentation loss and adversarial loss as objective. The proposed model has better segmentation performance than models without the adversarial loss.

Unpaired Data-to-Data Translation: Anomaly Detection

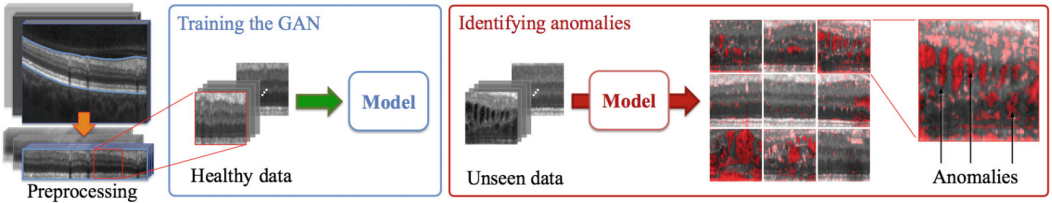


Figure: Schlegl et al. (2017) first trained unconditional image synthesis model with only healthy data, and then use it to calculate anomaly scores. Their anomaly scores are the summation of reconstruction error in raw data space and extracted feature space, respectively.

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3. Data-to-Data Translation

4. Other Advanced Topics

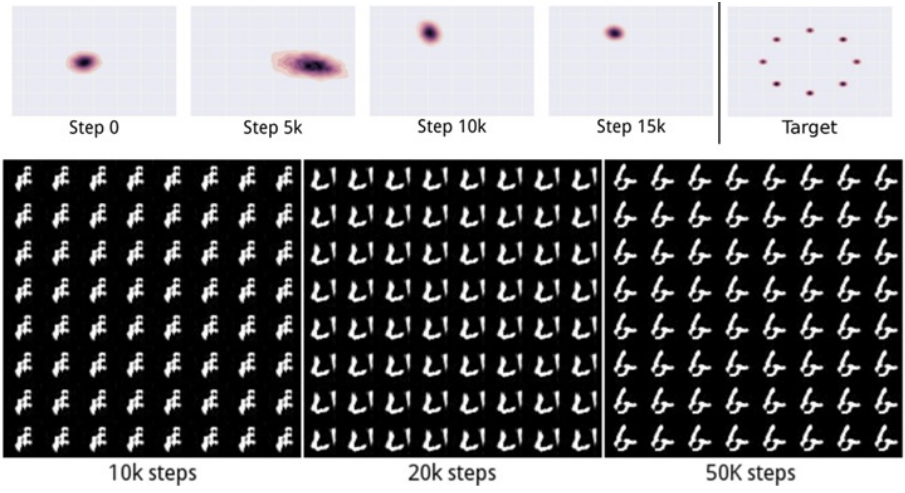
Mode Collapse

- Again, the adversarial learning process involves alternately maximizing and minimizing the value function:

$$\theta^* \in \arg\min_{\theta} \left(\max_{\phi} V(\theta, \phi) \right). \tag{26}$$

- This process seeks to find the best generator parameters θ while considering the optimal responses of the discriminator.
- For a given generator parameter θ, GANs approximate $\max_{\phi} V(\theta, \phi)$ by updating the discriminator parameter ϕ.
- However, there is usually a substantial gap between the theoretical maximum $\max_{\phi} V(\theta^{(t)}, \phi)$ and the value obtained at any particular iteration $V(\theta^{(t)}, \phi^{(t)})$.

Mode Collapse



Images are from Metz et al. (2017).

Mode Collapse: Unrolled GAN

- Unrolled GANs (Metz et al., 2017) aggressively update the discriminator parameters:

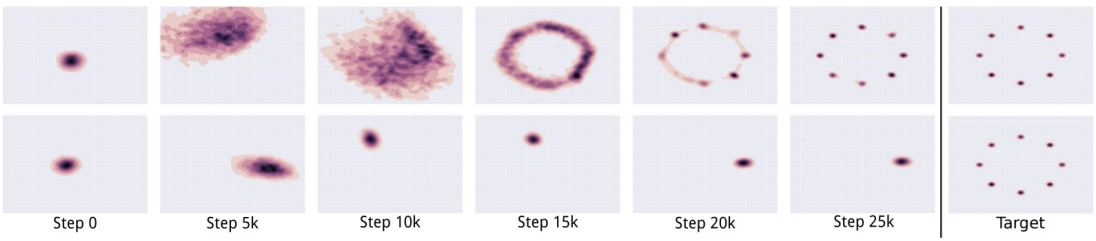
$$\phi^{(t,k+1)} = \phi^{(t,k)} + \eta^{(k)} \frac{dV(\theta^{(t)}, \phi)}{d\phi}|_{\phi=\phi^{(t,k)}} \tag{27}$$

  where $\eta^{(k)}$ represents the learning rate. By repeating this process K times, we may expect that $\phi^{(t,K)}$ better approximates the local maximum of $V(\theta^{(t)}, \phi)$.

- The generator parameter is then updated via:

$$\theta^{(t+1)} = \theta^{(t)} - \eta \frac{dV(\theta, \phi^{(t,K)})}{d\theta}|_{\theta=\theta^{(t)}} \tag{28}$$

Mode Collapse: Unrolled GAN



Images are from Metz et al. (2017).

Mode Collapse: Other Remedies

- Other remedies include modifying the value function.
- Popular approaches include Wasserstein generative models. These will be reviewed in detail in a later chapter on optimal transport-based approaches.

f-GAN

- We have reviewed the following relationship in GANs, which holds up to a constant addition and sign-preserving multiplication: Using optimal discriminators, we have:

$$\mathcal{D}_{\mathsf{JS}}(p \parallel p_\theta) \approx V(\theta, \hat{\phi}_n(\theta)).$$

- Nowozin et al. (2016) generalized the concept of using auxiliary networks to approximate other f-divergences:

$$\mathcal{D}_f(p||q) := \int_{\mathbb{R}^d} f\left(\frac{p(\boldsymbol{x})}{q(\boldsymbol{x})}\right) q(\boldsymbol{x}) d\boldsymbol{x} \tag{29}$$

  where $f : \mathbb{R}^+ \to \mathbb{R}$ is a convex function satisfying $f(1) = 0$.

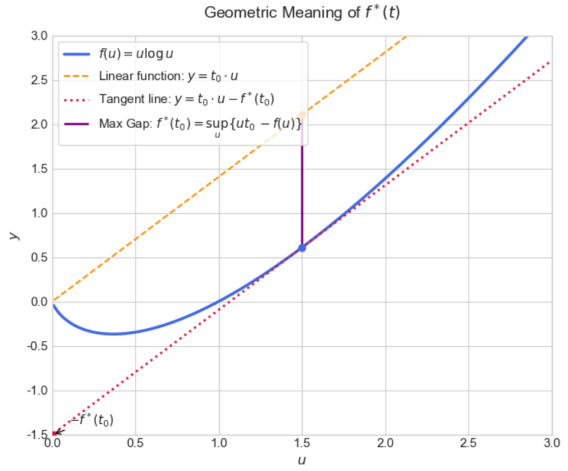
- The key idea is to introduce the convex conjugate function (or Fenchel conjugate, Hiriart-Urruty and Lemaréchal, 2004) to derive variational estimations of f-divergences.

f-Divergence

Name	$D_f(P\|Q)$	$f(u)$
Kullback-Leibler	$\int p(x) \log \frac{p(x)}{q(x)} \mathrm{d}x$	$u \log u$
Reverse KL	$\int q(x) \log \frac{q(x)}{p(x)} \mathrm{d}x$	$-\log u$
Pearson χ^2	$\int \frac{(q(x)-p(x))^2}{p(x)} \mathrm{d}x$	$(u-1)^2$
Squared Hellinger	$\int \left(\sqrt{p(x)} - \sqrt{q(x)}\right)^2 \mathrm{d}x$	$\left(\sqrt{u}-1\right)^2$
Jensen-Shannon	$\frac{1}{2}\int p(x) \log \frac{2p(x)}{p(x)+q(x)} + q(x) \log \frac{2q(x)}{p(x)+q(x)} \mathrm{d}x$	$-(u+1)\log \frac{1+u}{2} + u \log u$

List of popular examples of f-divergences, edited from Nowozin et al. (2016).

f-GAN



- We denote the convex conjugates of functions f by $f^*(t) := \sup_u \{ut - f(u)\}$.

f-GAN

- The f^* relates f and its subgradients. When f is convex and differentiable[3], the following properties hold:
  1. f^* is also convex and differentiable.
  2. Duality holds, i.e., $(f^*)^* = f$.
  3. The relation $f(u) + f^*(t) = ut$ holds if and only if $t = f'(u)$.
- When f' is invertible, $f^*(t) = (f')^{-1}(t)t - (f \circ f'^{-1})(t)$.

[3]For more general functions, check Hiriart-Urruty and Lemaréchal (2004).

f-GAN

Exercise 5

We assume that f is a convex function satisfying $f(1) = 0$ and is differentiable and assume that $\{T_\phi | \phi \in \Phi\}$ is sufficiently flexible such that for any given θ, there exists $\phi^*(\theta)$ where:

$$T_{\phi^*(\theta)}(\boldsymbol{x}) = f'\left(\frac{p(\boldsymbol{x})}{p_\theta(\boldsymbol{x})}\right). \tag{30}$$

Show that $\mathcal{D}_f(p||p_\theta) \geq \sup_{T_\phi}\Big(\int_{\mathbb{R}^d} T_\phi(\boldsymbol{x})p(\boldsymbol{x})d\boldsymbol{x} - \int_{\mathbb{R}^d} f^*(T_\phi(\boldsymbol{x}))p_\theta(\boldsymbol{x})d\boldsymbol{x}\Big)$, and the supreme is attained at $T_\phi = T_{\phi^*(\theta)}$.

f-GAN



f-GAN

- Define $F(\theta, \phi) := \int_{\mathbb{R}^d} T_\phi(\boldsymbol{x}) p(\boldsymbol{x})\, d\boldsymbol{x} - \int_{\mathbb{R}^d} f^*\big(T_\phi(\boldsymbol{x})\big) p_\theta(\boldsymbol{x})\, d\boldsymbol{x}$ and $\theta^* := \arg\min_\theta \big(\max_\phi F(\theta, \phi)\big)$.
- Then, $\max_\phi F(\theta, \phi) = F(\theta, \phi^*(\theta)) = \mathcal{D}_f(p \| p_\theta)$. Thus, p_{θ^*} is a minimizer of the f-divergence.
- **Example 1 (KL divergence)**: Let $f(u) = u \log u$ and $f^*(t) = \exp(t-1)$. We can express $F(\theta, \phi)$ as follows:

$$F(\theta, \phi) = \int_{\mathbb{R}^d} T_\phi(\boldsymbol{x}) p(\boldsymbol{x})\, d\boldsymbol{x} - \int_{\mathbb{R}^d} \exp\big(T_\phi(\boldsymbol{x}) - 1\big) p_\theta(\boldsymbol{x})\, d\boldsymbol{x}. \tag{31}$$

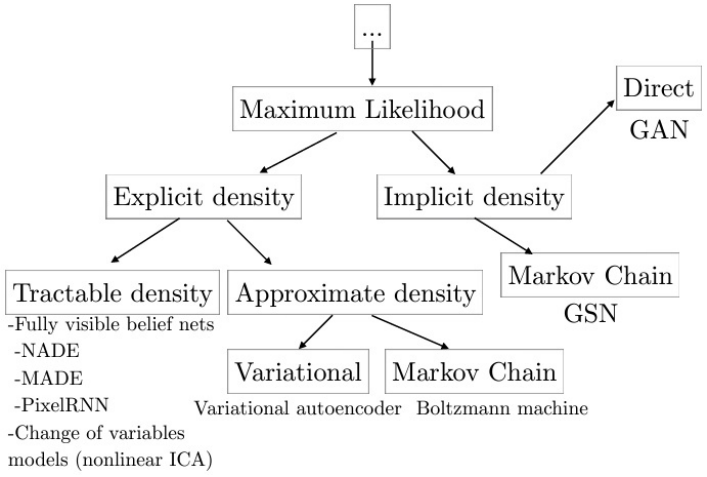
f-GAN

- **Example 2 (JS divergence)**: Let $f(u) = -(u+1)\log\frac{1+u}{2} + u\log u$ and $f^*(t) = -\log(2-\exp(t))$. We can express $F(\theta, \phi)$ as follows:

$$F(\theta,\phi) = \int_{\mathbb{R}^d} T_\phi(\boldsymbol{x}) p(\boldsymbol{x})\, d\boldsymbol{x} + \int_{\mathbb{R}^d} \log\Big(2 - \exp\big(T_\phi(\boldsymbol{x})\big)\Big) p_\theta(\boldsymbol{x})\, d\boldsymbol{x}. \tag{32}$$

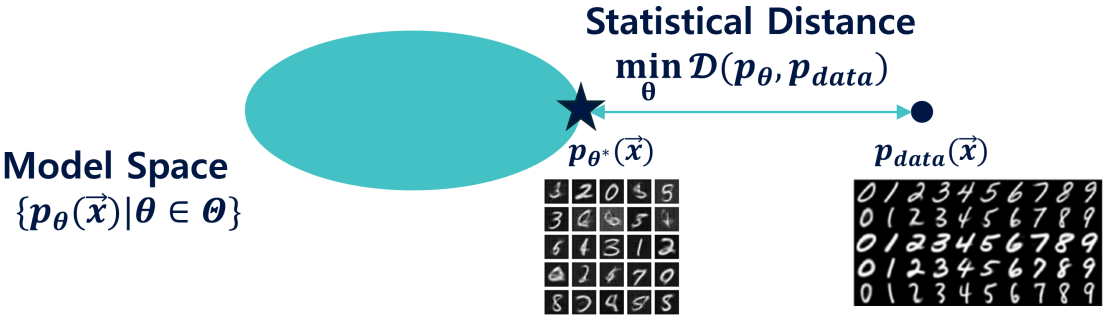
- GANs are special cases of f-GANs. When we model $T_\phi(\boldsymbol{x}) = \log D_\phi(\boldsymbol{x}) + \log 2$, $F(\theta,\phi) = V(\theta,\phi) + \log 4$, and $T_{\phi^*(\theta)}(\boldsymbol{x}) = \log D_{\phi^*(\theta)}(\boldsymbol{x}) + \log 2 = \log\frac{p(\boldsymbol{x})}{p(\boldsymbol{x})+p_\theta(\boldsymbol{x})} + \log 2$ hold.

Summary



The figure is from Goodfellow (2016).

Summary



Handwritten digit images are from Wolterink et al. (2017).

Summary

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1. Fundamentals of Probability Theory
2. Random Variables
3. Likelihood Maximization Principle
4. Basics of Probabilistic Modelling

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5. Autoregressive Model
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M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

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